

Application of Hybrid Recommendation System with Content-Based Filtering and Euclidean Distance for Lipstick Shade Recommendation

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Abstract

Selecting a lipstick shade that matches the natural characteristics of lip tone is generally conducted subjectively and conventionally, creating the need for a system capable of delivering automated, objective, and personalized recommendations. This study aims to design and implement a lip image-based lipstick shade recommendation system using a Deep Learning approach via the MobileNetV2 architecture combined with a Hybrid Recommendation System. The research pipeline comprises facial landmark detection using MediaPipe Face Mesh, lip area segmentation, feature extraction of average Red, Green, and Blue (RGB) values, automated labeling based on K-Means Clustering into Pinkish, Brownish, and Dark categories, lip type classification with MobileNetV2 through transfer learning, and lipstick shade recommendation through a combination of Rule-Based Filtering and Content-Based Filtering using Euclidean Distance. Evaluation results demonstrate that the MobileNetV2 model successfully classifies lip color types with a validation accuracy of 94.84%. In the recommendation system, evaluation on a commercial lipstick shade database successfully produced Top-3 recommendations with average similarity scores of 93.62% for the first recommendation, 92.26% for the second recommendation, and 91.41% for the third recommendation. Consequently, the developed system is proven accurate and effective in delivering lipstick shade recommendations tailored to the user's natural lip color characteristics.

Keywords: Deep Learning, MobileNetV2, Hybrid Recommendation System, Content-Based Filtering, Euclidean Distance, K-Means Clustering, Lipstick Shade Recommendation.

1. INTRODUCTION

The growth of the cosmetics industry in Indonesia has increased rapidly alongside heightened public awareness regarding self-appearance and personal grooming. Based on consumer market surveys in Indonesia, lip makeup products such as lipsticks, lip creams, and lip tints constitute the most widely used cosmetic category among women, with a usage prevalence reaching 80.4% [1]. Lipstick serves not only as a lip colorant to enhance facial aesthetics, but also as an essential element for expressing personal character and self-confidence.

Despite the widespread popularity of lipstick products, selecting the appropriate shade often remains a challenge for consumers. The baseline lip color characteristics of each individual are unique and highly variable. Scientific investigations by Vergnaud et al. have demonstrated that natural lip coloration is influenced by complex physiological determinants, including melanin concentration, hemoglobin perfusion, stratum corneum thickness, ethnicity, and chronological age [2], [3]. These baseline pigment variations cause an identical lipstick shade to yield substantially different visual results when applied across different individuals. This phenomenon is further reinforced by the findings of Charton et al., who established a multi-ethnic lip color chart and demonstrated the wide spectral diversity of human lip tones [4].

Most consumers continue to choose lipstick shades through conventional and subjective means, such as trying physical tester products at retail counters or relying on social media reviews. These conventional methods present hygiene

risks and often lead to color mismatches during online purchases. Several previous studies have investigated beauty product recommendation systems. Katariya and Patel designed a CNN model to recommend cosmetic products based on skin type and tone [5]. Furthermore, Gulati et al. developed the BeautifAI system, which offers personalized makeup recommendations based on facial attributes and occasion requirements [6]. Regarding recommendation computation, Pratama et al. applied the Content-Based Filtering method for cosmetic recommendation based on product attribute similarities [7]. Nevertheless, most existing studies continue to rely on general facial skin tones or catalog text descriptions as reference parameters. No prior research has specifically extracted and classified the natural lip color characteristics of users as the primary foundation for lipstick shade recommendations.

The application of Computer Vision and Deep Learning technologies based on Convolutional Neural Networks (CNN) has proven highly reliable in recognizing visual image patterns accurately [8], [9], [10]. One CNN architecture that is highly efficient for practical deployment is MobileNetV2, which employs inverted residuals and linear bottlenecks to maintain a lightweight computational footprint without sacrificing high predictive accuracy [11], [12]. To categorize lip color data lacking predefined ground-truth labels, the unsupervised K-Means Clustering algorithm can be utilized to partition data based on RGB color value similarities [13], [14].

In recommender systems, single-paradigm methods such as Rule-Based Filtering are effective for pruning candidates based on rigid category constraints, yet lack the flexibility needed to quantify continuous gradations of color proximity [15]. Conversely, Content-Based Filtering utilizing the Euclidean Distance geometric metric can measure numerical feature proximity in a continuous and precise manner [16]. Therefore, combining both approaches into a Hybrid Recommendation System offers an optimal solution. This study aims to design and implement a lip image-based lipstick shade recommendation system by employing MediaPipe Face Mesh [17] for automated lip segmentation, average RGB feature extraction, K-Means clustering, MobileNetV2 classification via transfer learning, and a combination of Rule-Based and Content-Based Filtering (Euclidean Distance) to generate precise Top-3 lipstick recommendations.

2. RESEARCH METHOD

2.1 Research Pipeline and System Architecture

This research was conducted through systematic phases encompassing data preprocessing, deep learning model training, hybrid recommendation system design, and system performance evaluation. The general pipeline of the lip image-based lipstick shade recommendation system is illustrated in Figure 1.

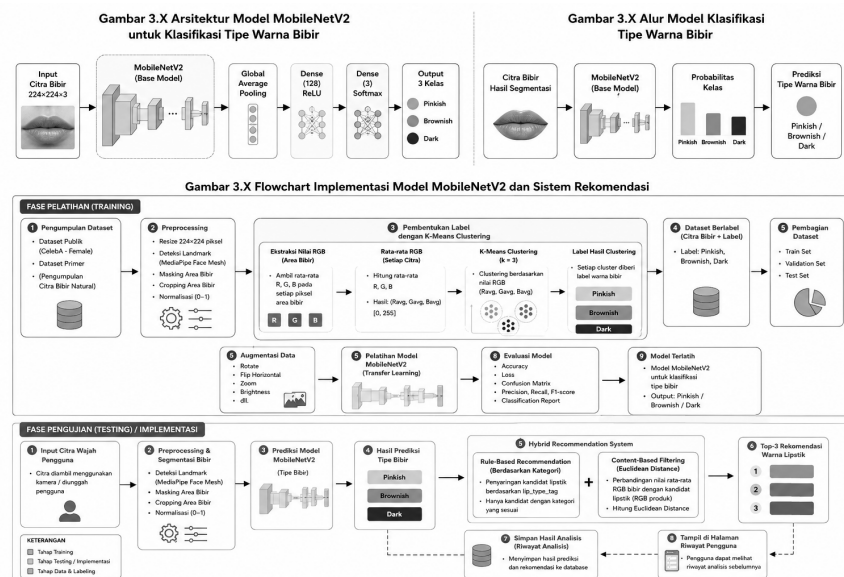


Figure 1: System Architecture and Flowchart of Lipstick Shade Recommendation Using MobileNetV2 and Hybrid Recommendation System

2.2 Dataset Collection and Partitioning

The dataset utilized in this research comprises a total of 18,100 facial images, consisting of:

1. Secondary Dataset (CelebA): 18,000 facial images sourced from the CelebFaces Attributes Dataset (CelebA) [18] featuring neutral expressions without heavy lip makeup.

2. Primary Dataset: 100 facial images of Indonesian women captured without lip makeup to enhance the representation of local lip pigmentation diversity.

The complete dataset was partitioned into a training set of 14,480 images (80%) and a validation set of 3,620 images (20%) for model training and hyperparameter evaluation.

2.3 Preprocessing and Lip Region Segmentation

The preprocessing stage aims to isolate the Region of Interest (ROI) corresponding to the lip area and standardize image dimensions:

1. Image Resizing: Standardizing input images to 224×224 pixels using OpenCV [19].
2. Facial Landmark Detection: Employing the MediaPipe Face Mesh framework to extract 468 3D facial landmark coordinates [17]. Outer and inner lip contours are precisely delineated (Figure 2(a)).
3. Segmentation and Masking: Constructing a binary polygon mask based on digital image segmentation principles [20], [21] derived from the lip boundary coordinates. All pixels outside the designated lip contour are set to zero (Figure 2(b)).
4. Normalization and ROI Cropping: Rescaling RGB pixel intensities to the $[0, 1]$ interval and cropping the image tightly bounded by the lip coordinates $(x_{\min}, y_{\min}, x_{\max}, y_{\max})$.



Figure 2: Stages of Facial Landmark Detection and User Lip Region Segmentation

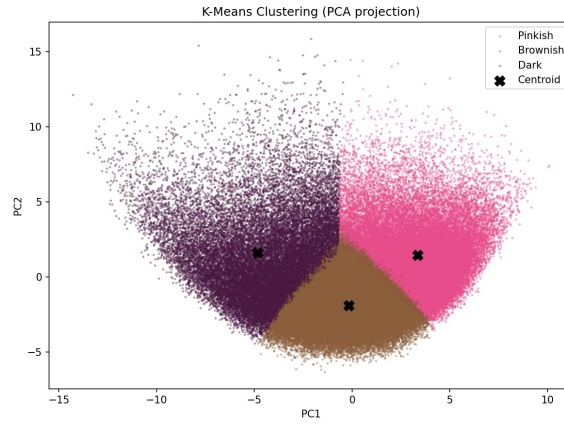
2.4 Color Feature Extraction and K-Means Clustering

Lip color characteristics are extracted from the mean intensities of the Red (R), Green (G), and Blue (B) color channels across all N segmented lip mask pixels using the formulation:

$$\bar{R} = \frac{1}{N} \sum_{i=1}^N R_i, \quad \bar{G} = \frac{1}{N} \sum_{i=1}^N G_i, \quad \bar{B} = \frac{1}{N} \sum_{i=1}^N B_i \quad (1)$$

These three scalar values $(\bar{R}, \bar{G}, \bar{B})$ serve as input features to the K-Means Clustering algorithm ($K = 3$, initialization k-means++, random state 42) to establish three distinct natural lip type clusters (Figure 3):

1. Pinkish: Bright tones with dominant reddish or soft pink hue components ($\bar{R} > 150, \bar{G} \approx 95 - 120, \bar{B} \approx 100 - 125$).
2. Brownish: Warm brownish tones ($\bar{R} \approx 125 - 155, \bar{G} \approx 75 - 100, \bar{B} \approx 65 - 90$).
3. Dark: Lip tones characterized by deep, concentrated pigmentation ($\bar{R} < 110, \bar{G} < 65, \bar{B} < 60$).

Figure 3: Distribution Visualization of Lip Color K-Means Clustering Results ($K = 3$)

2.5 MobileNetV2 Classification Model Training

The MobileNetV2 architecture was trained using a Transfer Learning approach initialized with pre-trained ImageNet weights [22], [23], supported by image data augmentation techniques to broaden feature variability [24]. Customized classification layers appended to the base feature extractor include: Global Average Pooling 2D, Dropout (0.3), Dense (128 units, ReLU activation), Dropout (0.2), and Dense Output (3 units, Softmax activation). The training hyperparameter configuration is summarized in Table 1.

Table 1: Hyperparameter Configuration for MobileNetV2 Model Training

Parameter	Value / Configuration
Base Architecture	MobileNetV2 (ImageNet Pre-trained)
Input Image Dimensions	$224 \times 224 \times 3$ pixels
Number of Epochs	30
Batch Size	32
Optimizer	Adam ($\beta_1 = 0.9, \beta_2 = 0.999, lr = 0.0001$)
Loss Function	Categorical Crossentropy
Number of Output Classes	3 (Pinkish, Brownish, Dark)

2.6 Hybrid Recommendation System

Once the MobileNetV2 classifier computes class probabilities for the user's lip tone ($\hat{y}_{user} \in \{\text{Pinkish, Brownish, Dark}\}$), the system identifies the most suitable lipstick shades through a two-stage hybrid mechanism:

2.6.1 Stage 1: Rule-Based Filtering

The system filters 281 commercial lipstick shades stored in the knowledge base by matching the `lip_type_tag` attribute with the user's predicted category:

$$\mathcal{C}_{\text{candidate}} = \{L_j \in \mathcal{D}_{\text{lipstick}} \mid \text{tag}(L_j) = \hat{y}_{user}\} \quad (2)$$

The resulting candidate pool comprises 147 products for Pinkish, 127 products for Brownish, and 14 products for Dark categories (Table 2).

Table 2: Rule-Based Recommendation Filtering Criteria Based on Lip Color Type

No	Predicted Lip Color Type	Knowledge Base Filtering Criteria	Candidate Product Count
1	Pinkish	<code>lip_type_tag</code> = Pinkish	147 products
2	Brownish	<code>lip_type_tag</code> = Brownish	127 products
3	Dark	<code>lip_type_tag</code> = Dark	14 products
Total Lipstick Database Entries			281 products

2.6.2 Stage 2: Content-Based Filtering Using Euclidean Distance

The system computes the geometric RGB color space distance (Euclidean Distance) between the user's lip color vector $\mathbf{u} = (R_u, G_u, B_u)$ and candidate lipstick shades $\mathbf{v}_j = (R_j, G_j, B_j)$ across all candidate items $L_j \in \mathcal{C}_{\text{candidate}}$:

$$d(\mathbf{u}, \mathbf{v}_j) = \sqrt{(R_u - R_j)^2 + (G_u - G_j)^2 + (B_u - B_j)^2} \quad (3)$$

The geometric distance is then mapped to a percentage similarity score relative to the maximum diagonal span of the RGB color cube $d_{\text{max}} = \sqrt{3} \times 255^2 \approx 441.673$:

$$\text{Similarity}(\mathbf{u}, \mathbf{v}_j) = \left(1 - \frac{d(\mathbf{u}, \mathbf{v}_j)}{d_{\text{max}}}\right) \times 100\% \quad (4)$$

The top three candidate products yielding the highest similarity scores are designated as the Top-3 Lipstick Recommendation (Algorithm 1).

Algorithm 1 Hybrid Recommendation System Algorithm for Lipstick Shade Recommendation

Require: User facial image I_{face} , Lipstick product database $\mathcal{D}_{\text{lipstick}}$, MobileNetV2 classification model $\mathcal{M}$.

Ensure: Top-3 lipstick shade recommendations along with calculated percentage similarity scores.

```

1:  $I_{\text{lip}} \leftarrow \text{SegmentLipRegion}(I_{\text{face}})$  ▷ MediaPipe Face Mesh segmentation
2:  $(R_u, G_u, B_u) \leftarrow \text{ExtractAverageRGB}(I_{\text{lip}})$ 
3:  $\hat{y}_{\text{user}} \leftarrow \mathcal{M}.\text{predict}(I_{\text{lip}})$  ▷ Classification: Pinkish, Brownish, or Dark
4:  $\mathcal{C}_{\text{candidate}} \leftarrow \emptyset$ 
5: for each  $L_j \in \mathcal{D}_{\text{lipstick}}$  do ▷ Stage 1: Rule-Based Filtering
6:   if  $L_j.\text{tag} == \hat{y}_{\text{user}}$  then
7:      $\mathcal{C}_{\text{candidate}} \leftarrow \mathcal{C}_{\text{candidate}} \cup \{L_j\}$ 
8:   end if
9: end for
10:  $\mathcal{R} \leftarrow []$ 
11: for each  $L_j \in \mathcal{C}_{\text{candidate}}$  do ▷ Stage 2: Content-Based Filtering
12:    $(R_j, G_j, B_j) \leftarrow L_j.\text{rgb}$ 
13:    $d_j \leftarrow \sqrt{(R_u - R_j)^2 + (G_u - G_j)^2 + (B_u - B_j)^2}$ 
14:    $S_j \leftarrow (1 - d_j / 441.673) \times 100\%$ 
15:   Append  $(L_j, d_j, S_j)$  to  $\mathcal{R}$ 
16: end for
17: Sort  $\mathcal{R}$  in ascending order by distance  $d_j$ 
18: return Top-3 elements of  $\mathcal{R}$ 

```

2.7 Performance Evaluation Metrics

Classification performance is evaluated using the Confusion Matrix and standard metrics: Accuracy, Precision, Recall, and F1-Score [25]:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

3. RESULTS AND DISCUSSION

3.1 MobileNetV2 Model Training and Evaluation Results

The MobileNetV2 architecture was trained for 30 epochs on the dataset annotated via K-Means clustering. Figure 4(a) depicts the convergence curves of accuracy and categorical loss, while Figure 4(b) displays the resulting Confusion Matrix evaluated on the validation dataset (803 samples).

The model attained stable convergence beyond the 15th epoch without indications of overfitting, yielding a final validation accuracy of 94.84% and a validation loss of 0.1612. Detailed performance metrics across individual classes are summarized in the Classification Report in Table 3.

The model demonstrates strong predictive performance on the Pinkish class (F1-score 97.6%). On the Brownish and Dark classes, F1-scores were 49.4% and 24.0%, respectively. These values reflect the predominance of lighter lip tones within the public facial dataset (class imbalance). Nevertheless, across all samples, the model achieved a weighted average F1-score of 93.8% and an overall accuracy of 94.0%, validating MobileNetV2's effectiveness in recognizing lip visual characteristics.

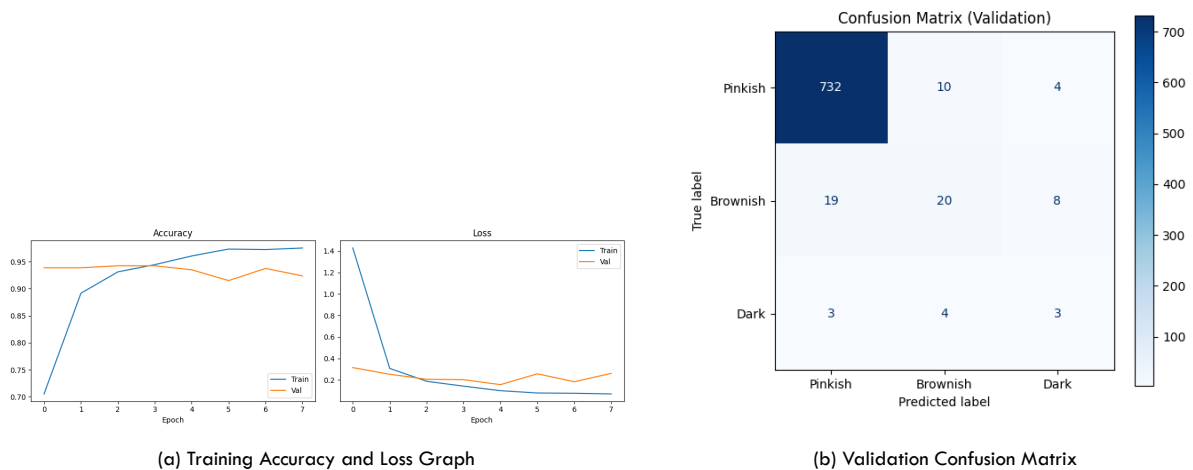


Figure 4: MobileNetV2 Performance Evaluation: (a) Accuracy and Loss Convergence Graph, (b) Confusion Matrix on Validation Data

Table 3: MobileNetV2 Model Performance Classification Report

Lip Tone Class	Precision	Recall	F1-Score	Support (Samples)
Pinkish	0.971 (97.1%)	0.981 (98.1%)	0.976 (97.6%)	746
Brownish	0.588 (58.8%)	0.426 (42.6%)	0.494 (49.4%)	47
Dark	0.200 (20.0%)	0.300 (30.0%)	0.240 (24.0%)	10
Overall Accuracy		0.940 (94.0%)		803
Macro Average	0.586	0.569	0.570	803
Weighted Average	0.939	0.940	0.938	803

3.2 Testing and Evaluation of the Hybrid Recommendation System

Evaluation of the recommendation system was conducted using 30 user test samples with diverse lip RGB color values. Selected sample test outputs are presented in Table 4, and the overall quantitative summary is detailed in Table 5.

Table 4: Sample Results of the Hybrid Recommendation System on User Test Data

No	User RGB	Lip Type	Shade Recommendation	Lipstick RGB	Euclidean (d)	Similarity (%)
1	(164, 121, 108)	Pinkish	1. Glossy Berry	(167, 97, 101)	25.18	94.3%
			2. Dusty Rose	(185, 115, 125)	27.68	93.7%
			3. Cozy Cherry	(173, 98, 94)	28.39	93.6%
2	(128, 102, 99)	Brownish	1. Chilled Raspberry	(134, 85, 109)	20.62	95.3%
			2. Chilled Cranberry	(132, 68, 82)	38.22	91.3%
			3. Matte Berry	(162, 83, 86)	41.06	90.7%
3	(145, 98, 55)	Pinkish	1. Satin Vermillion	(173, 95, 86)	41.88	90.5%
			2. Bold Ruby	(176, 88, 86)	44.97	89.8%
			3. Cozy Cherry	(173, 98, 94)	48.01	89.1%
4	(195, 167, 153)	Brownish	1. Mauve Bliss	(170, 110, 130)	66.36	85.0%
			2. Warm Terracotta	(180, 105, 95)	74.35	83.2%
			3. Rose Brown	(160, 95, 110)	81.12	81.6%

The performance summary demonstrates that the hybrid recommendation system consistently produces lipstick shade recommendations with high compatibility. The average color similarity for the primary recommendation (Top-1) reached 93.62%, with all Top-3 recommendations consistently exceeding 91%. The two-stage hybrid framework proves effective in filtering out mismatched categories during the Rule-Based phase while maximizing fine-grained color gradation matching during the Euclidean Distance stage.

3.3 Web-Based System User Interface Implementation

The system was deployed as a responsive web application. Users upload a facial photograph, after which the application automatically displays the segmented lip ROI, extracted RGB values, MobileNetV2 predicted classification, confidence score, and interactive Top-3 recommendation cards complete with color swatches and similarity percentages (Figure 5).

Table 5: Quantitative Summary of Similarity Scores for the Hybrid Recommendation System

Evaluation Metric	Quantitative Value
Number of Evaluation Test Samples	30 data points
Average Top-1 Similarity (First Recommendation)	93.62%
Average Top-2 Similarity (Second Recommendation)	92.26%
Average Top-3 Similarity (Third Recommendation)	91.41%
Highest Recorded Similarity Score	97.60%
Lowest Recorded Similarity Score	73.60%
Total Registered Commercial Lipstick Shades	281 shades

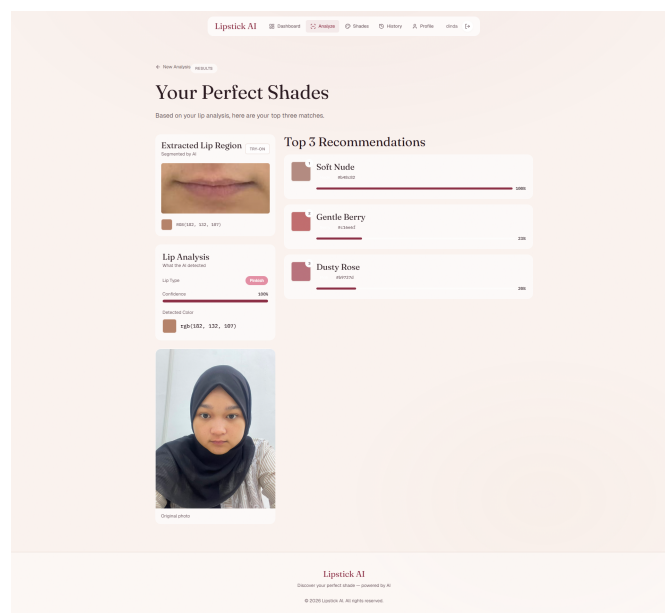


Figure 5: User Interface Implementation of Analysis Results and Lipstick Recommendations

3.4 Functional Testing (Black Box Testing)

Functional evaluation using Black Box Testing verified that all core software modules operate according to operational requirements, achieving a 100% PASS rate (Table 6).

Table 6: Black Box Testing Results for Analysis and Recommendation Modules

No	Test Scenario	Activity / Input	Expected Output	Status
1	Valid Image Upload	JPG/PNG facial image	Face detected and image preview rendered	PASS
2	Landmark Detection	Execute analysis	MediaPipe extracts 468 3D facial landmarks	PASS
3	Lip Segmentation	Delineate lip boundary	Polygon mask generated and lip region cropped	PASS
4	Color Classification	RGB extraction & inference	MobileNetV2 predicts class (Pinkish/Brownish/Dark)	PASS
5	Top-3 Recommendation	Rule filter & Euclidean metric	Top 3 lipstick shades with highest similarity displayed	PASS
6	Save History	Commit to database	Record persisted and retrievable via History menu	PASS

4. CONCLUSION AND FUTURE WORK

4.1 Conclusion

Based on the design, development, and evaluation conducted, this study has successfully established an automated, personalized lip image-based lipstick shade recommendation system integrating MediaPipe Face Mesh, MobileNetV2 via transfer learning, and a Hybrid Recommendation System. The MobileNetV2 model proved reliable in classifying natural lip tones into three defined categories (Pinkish, Brownish, and Dark), achieving a validation accuracy of 94.84% and a validation loss of 0.1612. Furthermore, integrating Rule-Based Filtering and Content-Based Filtering based on Euclidean Distance effectively optimized recommendation precision from a commercial cosmetic database, attaining average similarity scores of 93.62% for the first recommendation, 92.26% for the second recommendation, and 91.41% for the third recommendation, with a peak compatibility score of 97.60%. Functional validation through Black Box Testing verified that all application modules execute reliably with a 100% success rate.

4.2 Future Work

For future research and system enhancement, increasing primary sample diversity—particularly within the underrepresented Dark and Brownish pigmentation categories—is recommended. Incorporating specialized class imbalance countermeasures such as class weighting or SMOTE will further enhance minority class sensitivity. Additionally, subsequent developments may extend system functionality by introducing interactive Augmented Reality (AR) Virtual Try-On capabilities, as well as integrating skin undertone and occasion-based preferences to deliver even more comprehensive and adaptive cosmetic recommendations.

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